# Contents

# 1 Topological Space

## 1.1 Basic definition

**Definition 1.1.1**

Let $X$ be a non-empty set. A set $\mathcal{T}$ of subsets of $X$ is said to be a ***topology*** on $X$ if

(i) $X$ and the empty set, $\varnothing$, belong to $\mathcal{T}$,
(ii) the union of any (finite or infinite) number of sets in $\mathcal{T}$ belongs to $\mathcal{T}$, and
(iii) the intersection of any two sets in $\mathcal{T}$ belongs to $\mathcal{T}$.

The pair $(X, \mathcal{T})$ is called a ***topological space***.

**Definition 1.1.2**

Let $(X, \mathcal{T})$ be any topological space. Then the members of $\mathcal{T}$ are said to be ***open sets***.

**Proposition 1.1.3**

If $(X, \mathcal{T})$ is any topological space, then

(i) $X$ and $\varnothing$ are open sets,
(ii) the union of any (finite or infinite) number of open sets is an open set,
(iii) the intersection of any finite number of open sets is an open set.

**Definition 1.1.4**

Let $(X, \mathcal{T})$ be a topological space. A subset $S$ of $X$ is said to be a ***closed set*** in $(X, \mathcal{T})$ if its complement in $X$, namely $X \setminus S$, is open in $(X, \mathcal{T})$.

**Proposition 1.1.5**

If $(X, \mathcal{T})$ is any topological space, then

(i) $\varnothing$ and $X$ are closed sets,
(ii) the intersection of any (finite or infinite) number of closed sets is a

closed set and
(iii) the union of any finite number of closed sets is a closed set.

**Definition 1.1.6**

A subset $\mathcal{S}$ of a topological space $(X, \mathcal{T})$ is said to be ***clopen*** if it is both open and closed in $(X, \mathcal{T})$.

## 1.2 Basis for a Topology

**Definition 1.2.1**

Let $(X, \mathcal{T})$ be a topological space. A collection $\mathcal{B}$ of open subsets of $X$ is said to be a ***basis*** for the topology $\mathcal{T}$ if every open set is a union of members of $\mathcal{B}$.

*Remark.*
拓扑基不是唯一的，根据定义，一旦一个集族 $\mathcal{B}$ 成为了一个拓扑基，我们无论往 $\mathcal{B}$ 添加多少开集，它还是这个拓扑的基. 因此直觉上，拓扑的基是越小越好用的.

*Note.*
也就是说，拓扑基是通过任意并运算生成拓扑的对象.

**Proposition 1.2.2**

Let $X$ be a non-empty set and let $\mathcal{B}$ be a collection of subsets of $X$. Then $\mathcal{B}$ is a basis for a topology on $X$ if and only if $\mathcal{B}$ has the following properties:
(a) $X = \bigcup_{B \in \mathcal{B}} B$, and
(b) for any $B_1, B_2 \in \mathcal{B}$, the set $B_1 \cap B_2$ is a union of members of $\mathcal{B}$.

*Remark.*
这里不谈与 $X$ 上的任何一个具体的拓扑的关系，只谈一个集族什么时候能够生成一个拓扑.

> *Note.*
> 这个命题刻画了一个集族距离成为一组基有多远．即一个集族需要什么额外条件，才能通过任意并这个运算生成出一个拓扑．命题中的条件 (a) 无非对应于 Definition 1.1.1 (i)．而通过基任意并生成的这个操作，天然就是为了 Definition 1.1.1 (ii) 而建立的．因此 (b) 就是为了满足1.1.1 (iii) 而提出的条件．我们发现生成拓扑对于有限交封闭的要求，完全地变成了对于基的有限交的封闭的要求．

*Proof.* If $\mathcal{B}$ is a basis for a topology on $X$, say $\mathcal{T}_{\mathcal{B}}$. Then $X \in \mathcal{T}$ means that there is a collection of sets in $\mathcal{B}$ ,say $\mathcal{B}_1$, such that

$$X = \bigcup_{B \in \mathcal{B}_i} B \subseteq \bigcup_{B \in \mathcal{B}} B$$

And since every sets in $\mathcal{B}$ is a subset of $X$, it follows that (a) holds. To show (b), we only need to see that $\mathcal{B} \subseteq \mathcal{T}_{\mathcal{B}}$. Since $\mathcal{T}$ is closed under finite intersection, we have $B_1 \cap B_2 \in \mathcal{T}_{\mathcal{B}}$ .

Then conversely , if $\mathcal{B}$ has the properties (a), (b), we only show that the collection $\langle \mathcal{B} \rangle$ of union of the members in $\mathcal{B}$, satisfies Definition 1.1.1 (iii). We take $A_1, A_2 \in \langle \mathcal{B} \rangle$, and suppose that

$$A_i = \bigcup_{j \in I_i} B_j, \quad B_j \in \mathcal{B}, \quad I_i \text{ is a index set }, i = 1, 2$$

Then

$$\begin{aligned} A_1 \cap A_2 &= \left( \bigcup_{j_1 \in I_1} B_{j_1} \right) \cap \left( \bigcup_{j_2 \in I_2} B_{j_2} \right) \\ &= \bigcup_{j_2 \in I_2} \left( \left( \bigcup_{j_1 \in I_1} B_{j_1} \right) \cap B_{j_2} \right) \\ &= \bigcup_{j_2 \in I_2} \left( \bigcup_{j_1 \in I_1} (B_{j_1} \cap B_{j_2}) \right) \\ &= \bigcup_{j_1 \in I_1, j_2 \in I_2} B_{j_1} \cap B_{j_2} \end{aligned}$$

Since every $B_{j_1} \cap B_{j_2}$ is a union of members of $\mathcal{B}$, and $A_1 \cap A_2$ is a union of such $B_{j_1} \cap B_{j_2}$, then $A_1 \cap A_2$ is a union of members of $\mathcal{B}$. □

**Proposition 1.2.3**

Let $(X, \mathcal{T})$ be a topological space. A family $\mathcal{B}$ of open subsets of $X$ is a basis for $\mathcal{T}$ if and only if for any point $x$ belonging to any open set $U$, there is a $B \in \mathcal{B}$ such that $x \in B \subseteq U$.

*Remark.*
表述 "there is a $B \in \mathcal{B}$ such that $x \in B \subseteq U$ " 就是在说 "$U$ 可以由 $\mathcal{B}$ 通过 '并' 生成". 带入这句话之下, 这个命题几乎就是一句 "废话". 同样的道理也出现在 Proposition 1.2.4

*Note.*
想要看 $\mathcal{B}$ 是否作为基生成了拓扑 $\mathcal{T}$ . 我们不用谈 $\mathcal{B}$ 中的元素如何如何 "并" 出每一个开集 $U$. 而是只需要 $\mathcal{B}$ 对于开集是足够 "精细" 的, 以至于对于任意的开集 $U$, 我们可以用 $\mathcal{B}$ 里面的元素在 $U$ 上 "框出" 它的任意一个点.

*Proof.* If $\mathcal{B}$ is a basis for $\mathcal{T}$. Then we can write $U$ as

$$U = \bigcup_{i \in I} B_i$$

Then for any $x \in U$, there exists $i_0 \in I$ such that $x \in B_{i_0} \subseteq U$.

Now we show the converse. First we show that $\mathcal{B}$ is a basis . We use Proposition 1.2.2 to verify this. Since $X$ is a open set , for any $x \in X$, there is a $B_x \in \mathcal{B}$ , such that $x \in B_x \subseteq X$, we get

$$X = \bigcup_{x \in X} B_x \subseteq \bigcup_{B \in \mathcal{B}} B \subseteq X$$

Thus

$$X = \bigcup_{B \in \mathcal{B}} B$$

Take any $B_1, B_2 \in \mathcal{B}$, we know $B_1 \cap B_2$ is again an open set. Then similarly, we get

$$B_1 \cap B_2 = \bigcup_{x \in B_1 \cap B_2} B_x$$

is a union of members in $\mathcal{B}$. Thus $\mathcal{B}$ is actually a basis. Finally , we only need to show that $\mathcal{B}$ generates $\mathcal{T}$ , that is $\langle \mathcal{B} \rangle = \mathcal{T}$. $\langle \mathcal{B} \rangle$ is the collection of unions of members in $\mathcal{B}$. $\langle\mathcal{B}\rangle \subseteq \mathcal{T}$ is obvious. $\mathcal{T} \subseteq \langle\mathcal{B}\rangle$ concludes from

$$U = \bigcup_{x\in U} B_x \in \langle\mathcal{B}\rangle$$

$\square$

**Proposition 1.2.4**

Let $\mathcal{B}$ be a basis for a topology $\mathcal{T}$ on a set $X$. Then a subset $U$ of $X$ is open if and only if for each $x \in U$ there exists a $B \in \mathcal{B}$ such that $x \in B \subseteq U$.

*Note.*
把基 $\mathcal{B}$ 看成是 $\mathcal{T}$ 中代表着 “足够精细” 的某一批. 验证 $U$ 是否是开集, 只需要看 $U$ 是否可以被这一小批集合 “概括”.

*Proof.* $\implies$ is immediately from Proposition 1.2.3. Now suppose that for each $x \in U$, ther exists a $B_x \in \mathcal{B}$ such that $x \in B \subseteq U$, then

$$U = \bigcup_{x\in U} B_x \in \langle\mathcal{B}\rangle = \mathcal{T}$$

$\square$

**Proposition 1.2.5**

Let $\mathcal{B}_1$ and $\mathcal{B}_2$ be bases for topologies $\mathcal{T}_1$ and $\mathcal{T}_2$, respectively, on a non-empty set $X$. Then $\mathcal{T}_1 = \mathcal{T}_2$ if and only if

1. for each $B \in \mathcal{B}_1$ and each $x \in B$, there exists a $B' \in \mathcal{B}_2$ such that $x \in B' \subseteq B$, and
2. for each $B \in \mathcal{B}_2$ and each $x \in B$, there exists a $B' \in \mathcal{B}_1$ such that $x \in B' \subseteq B$.

*Proof sketch.*
考虑这句话: 表述 “there is a $B \in \mathcal{B}$ such that $x \in B \subseteq U$ ” 就是在说 “$U$ 可以由 $\mathcal{B}$ 通过 ‘并’ 生成”. 1. 是说 $\mathcal{B}_1$ 中任意集合可以由 $\mathcal{B}_2$“并” 出来. 自然地 $\mathcal{T}_1$ 中任意集合也被 $\mathcal{B}_2$“并出来”. 因此 1. 就是 $\mathcal{T}_1 \subseteq \mathcal{T}_2$. 同样地, 2. 就是 $\mathcal{T}_2 \subseteq \mathcal{T}_1$.